

OpenCap Portable Pro: Camera Setup & Calibration Guide

Accurate biomechanical analysis relies entirely on precise camera calibration. This guide outlines the required steps to configure your iPhones and record the necessary Intrinsic and Extrinsic calibration videos.

Phase 1: Camera App Setup

To ensure consistent frame rates and prevent the camera from auto-adjusting during a trial, we recommend use of the Blackmagic Camera app.

1. Download the App

1. Open the App Store on all iPhones used for data collection.
2. Search for and download Blackmagic Camera (Free).

2. Configure Camera Settings

Open the app and navigate to the Settings gear icon. Apply the following configurations to *every* phone:

- Resolution: 720p
- Frame Rate (FPS): 60 fps
- Codec: H.264 or HEVC (H.265)
- Lens: Select the Main (1x) Wide lens. *Do not use the Ultra-Wide (0.5x) or Telephoto lenses.*

3. Lock Exposure and Focus

During recording, the camera must not change focus or brightness.

1. Manually set the focus slider close to the maximum. We recommend 0.8 for most trials. This must be set the same for the entire trial.
2. Lock the Focus and Exposure by tapping the lock icons on the screen (they should turn red or indicate they are locked).
3. Shutter Speed: Manually set your shutter speed to 1/120 or faster to minimize motion blur during dynamic movements.
4. Turn off all image stabilization.

Phase 2: Recording "Intrinsic" Videos

Purpose: To map the unique lens distortion of each individual camera.

Note: You only need to do this once per device unless the camera is physically damaged or you change the focus or recording resolution.

Execution Steps

1. Open the Blackmagic app on Cam0.
2. Have a researcher hold the printed OpenCap checkerboard. Ensure the board is flat and rigid (mounted to a hard surface).
3. Start the recording.
4. Move the checkerboard through the camera FOV: The assistant must slowly move the checkerboard throughout the camera's entire field of view for about 15 to 30 seconds:

- Move it close to the camera, then far away.
 - Move it to all four corners of the frame.
 - Slowly tilt the board slightly up, down, left, and right (approx. 20-30 degrees) so the camera sees it from different angles.
 - Preferably record more than one trial.
 - Ensure the checkerboard does not leave the capture volume during the recording.
5. Stop Recording and repeat this exact process individually for Cam1, Cam2, etc.
 6. Naming: When importing to the OpenCap Launcher, these will be assigned to the Intrinsic trial type.

Phase 3: Recording "Extrinsics" (Spatial calibration) Videos

Purpose: To teach the software exactly where the cameras are positioned in 3D space relative to each other and to establish the floor level.

Note: You must record a new Extrinsics video every time you move the tripods or start a new session.

Execution Steps

1. **Position the Cameras:** Set up your iPhones on their tripods around the capture volume. Ensure they are secure and will not be bumped.
2. **Place the Checkerboard:** * Place the checkerboard in the exact center of the room where the subject will be performing the trials.
 - **Vertical (Wall/Stand):** Prop the board up perfectly straight so it aligns with gravity.
 - **Horizontal (Floor):** Lay the board perfectly flat on the ground (ideal for 360-degree camera setups).
 - *Note: Remember which placement you used, as you will need to select it in the GUI later.*
3. **Verify Visibility:** Look at the screen of every iPhone. **Every camera must be able to see the entire checkerboard clearly at the same time.** No corners can be cut off.
4. **Record:** * Start recording on all cameras simultaneously.
 - Leave the checkerboard completely still for **3 to 5 seconds**.
 - Stop recording on all cameras.
5. **Naming:** When importing to the OpenCap Launcher, these videos will be assigned to the Calibration trial type.

Best Practices & Troubleshooting

- **Lighting:** Ensure the room is brightly lit. Shadows across the checkerboard can cause the detection algorithm to fail.
- **Rigidity:** If your printed checkerboard bends or warps, the calibration will fail. Glue or tape it to a piece of wood, acrylic, or thick cardboard.

- **No Auto-Focus:** If the checkerboard looks blurry, your focus was not locked correctly. Re-tap to focus and lock it before recording.